

Kaggle #09



What is Feature engineering? An Overview

- Transforming raw data into informative features
- We are making the inputs more useful for models
- Helps models to understand patterns more easily
- Often has a bigger impact than choosing a complex model

Why Feature Engineering Matters

- Real-world data rarely arrives in model ready form
- Good features simplify the learning problem
- We need to reduce noise and highlight structure
- Small feature changes can produce large accuracy gains
- A basic example:

Changing **Fare** to **Fare per Person** gives us greater predictive power

Where does it fit in?

- Data collection
- Preprocessing (cleaning, missing values, types, scaling)
- **Feature Engineering**
- Model training
- Evaluation and iteration

The 4 parts of Feature Engineering

Different sources define feature engineering differently, but in practice it breaks down into:

- Feature Creation
- Feature Encoding
- Feature Selection
- Feature Extraction

Feature Creation (Overview)

- Creating new features from existing data
- Makes relationships clearer to the model
- Common techniques:
 - Transformations (log, sqrt, scaling)
 - Interactions (ratios, products)
 - Aggregations (group statistics)
 - Binning (discretising continuous values)
 - Domain-specific features (dates, text length, distances)
 - Scaling (*when it produces a new informative version)

Feature Creation - some basic examples

- BMI = $\text{weight} / \text{height}^2$
- Fare per Person = $\text{total_fare} / \text{group_size}$
- Rooms per Household = $\text{rooms} / \text{households}$
- Extracting date components: year, month, day, hour
- Text length = number of characters or words

Feature Encoding (Overview)

- Models require numeric inputs
- Responsible for converting categorical features
- Common encoding methods:
 - One-hot encoding (low-cardinality categories)
 - Ordinal encoding (ordered categories)
 - Target encoding (large categories, requires CV)
- Choice of encoding depends on model and data size

Feature Selection (Overview)

- Reduces noise and avoids redundant features
- Helps prevent overfitting
- Speeds up training and simplifies models
- Common approaches:
 - Filter methods (correlation, chi-square)
 - Embedded methods (LASSO, tree feature importance)
 - Wrapper methods (RFE)

Feature Extraction (Overview)

- Creating new, compact representations of the data
- Reduces dimensionality while preserving important information
- Useful when features are high-dimensional, noisy, or strongly correlated
- **Common techniques (includes dimensionality reduction):**
 - PCA (Principal Component Analysis) - linear dimensionality reduction
 - SVD (Singular Value Decomposition) - behind PCA and LSA
 - Autoencoders — nonlinear dimensionality reduction
 - Text embeddings (TF-IDF, Word2Vec, BERT)
 - Image embeddings (CNN feature vectors)
 - t-SNE / UMAP (mainly for visualisation)

New Kaggle Competition! (December 2025)

The screenshot shows the Kaggle interface for the 'Diabetes Prediction Challenge'. On the left is a sidebar with navigation links: Create, Home, Competitions (highlighted), Datasets, Models, Benchmarks, Game Arena, Code, Discussions, Learn, and More. The main content area features the competition title 'Diabetes Prediction Challenge' and subtitle 'Playground Series - Season 5, Episode 12'. Below this is a horizontal menu with tabs: Overview (selected), Data, Code, Models, Discussion, Leaderboard, Rules, Team, and Submissions. The 'Overview' section contains a welcome message and the goal: 'Your Goal: Predict the probability that a patient will be diagnosed with diabetes.' On the right, there is a 'Submit Prediction' button, an illustration of two ducks in a medical setting, and information about the competition host (Kaggle), prizes (Swag), and participation rules.

Diabetes Prediction Challenge
Playground Series - Season 5, Episode 12

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

Welcome to the 2025 Kaggle Playground Series! We plan to continue in the spirit of previous playgrounds, providing interesting and approachable datasets for our community to practice their machine learning skills, and anticipate a competition each month.

Your Goal: Predict the probability that a patient will be diagnosed with diabetes.

Competition Host
Kaggle

Prizes & Awards
Swag
Does not award Points or Medals

Participation

- Target variable: `diagnosed_diabetes` (0 or 1)
- You must output a probability, not a class label
- Evaluation metric: AUC-ROC
- Measures how well predicted probabilities rank positive vs negative cases

TLDR

- Feature engineering transforms raw data into useful inputs for models
- It often improves performance more than changing the model itself
- It fits between preprocessing and model training in the ML pipeline
- The four parts of Feature Engineering:
 - Feature Creation - make new features (transformations, interactions, domain features)
 - Feature Encoding - turn categories into numeric form
 - Feature Selection - remove noisy or redundant features
 - Feature Extraction - create compact representations
- Strong features reduce noise, highlight structure, and help models learn patterns effectively

Your feedback matters to us!

